SYSTEM DESIGN DOCUMENTATION

for

MedTrack

Version 1.0

Prepared by

Group No :19

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Course: **CS2011E Database Management Systems**

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1. PURPOSE

This system design document for the MedTrack Database Management System outlines a specialized data storage and retrieval framework. The framework is derived from a thorough examination of the hospital system's functional requirements through internet and other sources of information, with a focus on preserving patient data accuracy, security, and other things. By scrutinizing the system's use case scenarios, an Entity-Relationship model has been developed that accurately represents the medical data structure. This model has been subsequently transformed into a relational schema, tailored to align with the chosen Database Management System (DBMS) platform to be utilised.

1.1 Document Objectives

This document outlines the specifications, functionalities, and design decisions taken or taking for MedTrack. The system is designed to improve hospital operations by streamlining workflows (such as patient registration, appointment scheduling, and laboratory tracking), reducing medical errors, minimizing administrative overhead, and enhancing the overall patient experience. The document serves as a guide for developers, stakeholders, and healthcare administrators to understand system capabilities and implementation strategies etc.

1.2 Intended Audience and Document Overview

The intended audience includes:

- **Healthcare Administrators** – To understand how MedTrack can aid in efficient hospital management and statistical reporting etc.
- **Medical Staff (Doctors, Pharmacists, Lab Technicians)** – To understand their specific workflows, such as diagnosing, dispensing, or uploading reports etc.
- **Receptionists/Front Desk** – To understand appointment and patient registration handling.
- **Developers** – To gain technical insights into the backend functionality, database schema, and security implementations.
- **Project Managers** – To oversee the development process and ensure healthcare for every patient and project goals are met.
- **Testers** – To validate the functionality of the software and review data integrity.

This document covers the following:

- System objectives and functionalities.
- Assumptions and constraints.
- Database-wide design decisions, including behaviour, security, and performance considerations taken.
- Data administrative functions, including ER models and relational schemas.

1.3 Definitions, Acronyms, and Abbreviations

SL.NO	ABBREVIATIONS	DEFINITIONS
1.	**DBMS**	Database Management System
2.	**EHR/EMR**	Electronic Health Record / Electronic Medical Record
3.	**ADMIN**	Healthcare administrator with access to all staff and system functionalities
4.	**STAFF**	Medical and administrative employees (Doctors, Receptionists, Pharmacists, Lab Techs)
5.	**API**	Application Programming Interface
6.	**JWT**	JSON Web Token (used for secure authentication)
7.	**ERD**	Entity-Relationship Diagram
8.	**SQL**	Structured Query Language

1.4 References

- Elmasri, R., & Navathe, S. B. (2015). Fundamentals of Database Systems (7th ed.). Pearson.
- Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). Database System Concepts (7th ed.). McGraw-Hill Education.
- PostgreSQL Global Development Group. (2024). PostgreSQL Documentation.
- Medical operational data privacy best practices (HIPAA general guidelines).

2. ASSUMPTIONS AND CONSTRAINTS

2.1 Assumptions

- The system will be used by registered hospital staff and registered patients.
- The hospital will follow a structured process for patient registration, appointment booking, lab testing, and pharmacy inventory management.
- Stable internet or hospital intranet connectivity is required for real-time operations, electronic medical record access, and database updates.
- Users will have basic knowledge of operating a digital web interface.

2.2 Constraints

- Only authorized hospital staff and authenticated patients have permission to access the web application no other authorization will be permitted.
- Administrators have access to: Manage all staff and doctor profiles, manage system configurations, and generate system-wide reports.
- Doctors have access to: View patient history, manage their daily schedules, record diagnoses, order lab tests, and create digital prescriptions.
- Receptionists have access to: Register patients, schedule/modify appointments, and view doctor availability.
- Pharmacists have access to: View active prescriptions, dispense medicines, and manage pharmacy inventory stock levels.
- Lab Technicians have access to: View pending lab test requests, enter test values, and upload digital reports.
- Patients have access to: Book/cancel appointments, update contact info, and view their own medical history and prescriptions.
- Security protocols (Role-Based Access Control and JWT) must be strictly implemented to prevent unauthorized access to sensitive medical records to not cause any unwanted breach.
- The system must support concurrent users (e.g., multiple receptionists booking appointments simultaneously) without performance delay and degradation.

3. DATABASE – WIDE DESIGN DECISIONS

3.1 BEHAVIOUR

3.1.1 User Authentication (Login/Registration)

- All users of the MedTrack system that is — Administrators, Doctors, Receptionists, Pharmacists, Lab Technicians, and Patients—must authenticate themselves through a secure login mechanism.
- Registered users can log in using their unique ID and Password.
- User credentials are securely stored and managed within the database.
- The system validates user credentials before granting access to any functionality.
- In case of invalid login attempts, the system displays an appropriate error message “Invalid Credentials”.
- Upon successful authentication, users are redirected to their dashboard.

3.1.2 Role-Based Access Behaviour

The system strictly follows a Role-Based Access Control (RBAC) model to ensure that each user interacts only with functionalities relevant to their responsibilities.

Administrator

- Full control over system operations
- Manages staff and doctor profiles
- Configures system settings and accesses reports

Doctor

- Accesses patient medical history and records
- Records diagnoses and treatment details
- Generates prescriptions and orders lab tests

Receptionist

- Registers new patients
- Schedules, updates, and cancels appointments
- Monitors doctor availability
- Pharmacist
- Views doctor-issued prescriptions
- Dispenses medicines
- Updates and maintains inventory records

Lab Technician

- Views pending lab test requests
- Enters test results and uploads reports
- Updates lab test status

Patient

- Manages personal profile information
- Books, reschedules, or cancels appointments
- Views prescriptions and personal medical records

This role-based structure ensures data privacy, prevents unauthorized access, and maintains system integrity.

3.1.3 Core System Behaviour and Operations

The MedTrack system supports a set of integrated operations that enable efficient hospital management:

- Patient registration and maintenance of electronic medical records
- Management of doctor and staff information
- Appointment scheduling, tracking, and updates
- Prescription generation and historical record keeping
- Laboratory test ordering, processing, and report storage
- Pharmacy inventory tracking and stock updates

All operations are performed in real-time, ensuring that the database remains consistent and up-to-date across all user interfaces. The system is designed to minimize redundancy, maintain data accuracy, and support seamless coordination between different hospital departments.

3.2 DBMS Platform

The MedTrack system is implemented using a modern web-based technology stack:

Database: MySQL

Used for storing and managing structured healthcare data, including patient records, appointments, prescriptions, and inventory. It ensures efficient querying, data consistency, and relational integrity.

Backend: Node.js with Express.js

Responsible for handling server-side logic, API endpoints, authentication using JWT, and communication with the database. Express.js provides a flexible and scalable framework for building RESTful services.

Frontend: Next.js

Provides a responsive and user-friendly interface for all user roles and communicates with the backend through secure API calls.

3.3 Security Requirements

Given the sensitive nature of medical data, the system incorporates multiple layers of security:

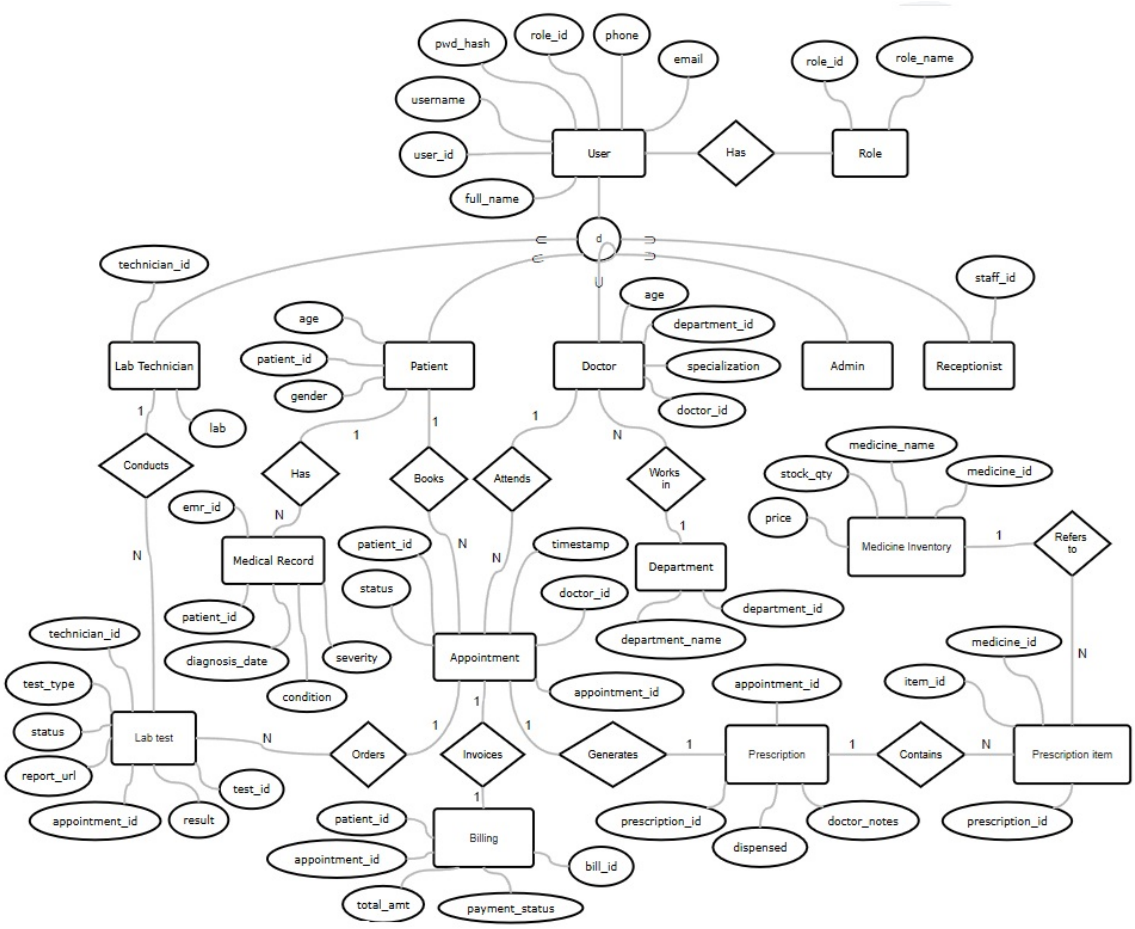
- Implementation of Role-Based Access Control (RBAC) to restrict access based on user roles
- Secure authentication using JSON Web Token
- Passwords stored in encrypted that is the hashed format.
- Secure communication using HTTPS to prevent data interception
- Regular database backups to prevent data loss and ensure recovery

3.4 Performance and Availability Decisions

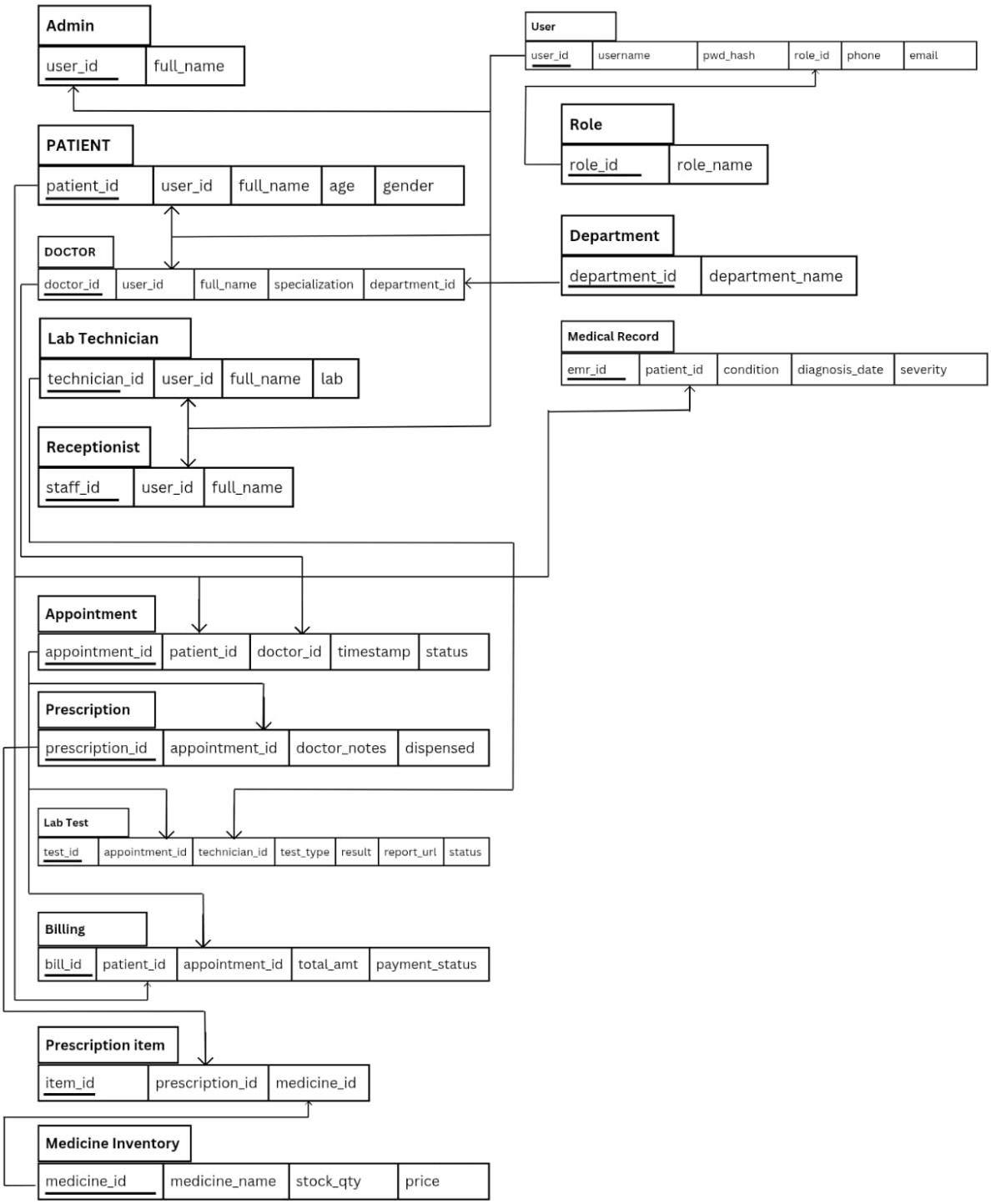
- The system is designed to ensure efficient performance and continuous availability:
- Most user operations such as login, appointment booking, and data retrieval are processed within a few seconds
- Real-time updates are supported for appointments, prescriptions, and inventory changes
- The system is capable of handling multiple concurrent users without performance degradation
- Database queries are optimized to reduce response time and improve efficiency
- The architecture is scalable to accommodate increasing data volume and user load
- Slight delays may occur for complex operations such as analytics and report generation, but overall system responsiveness is maintained

4. DATA ADMINISTRATIVE FUNCTIONS

4.1 Entity-Relation Model



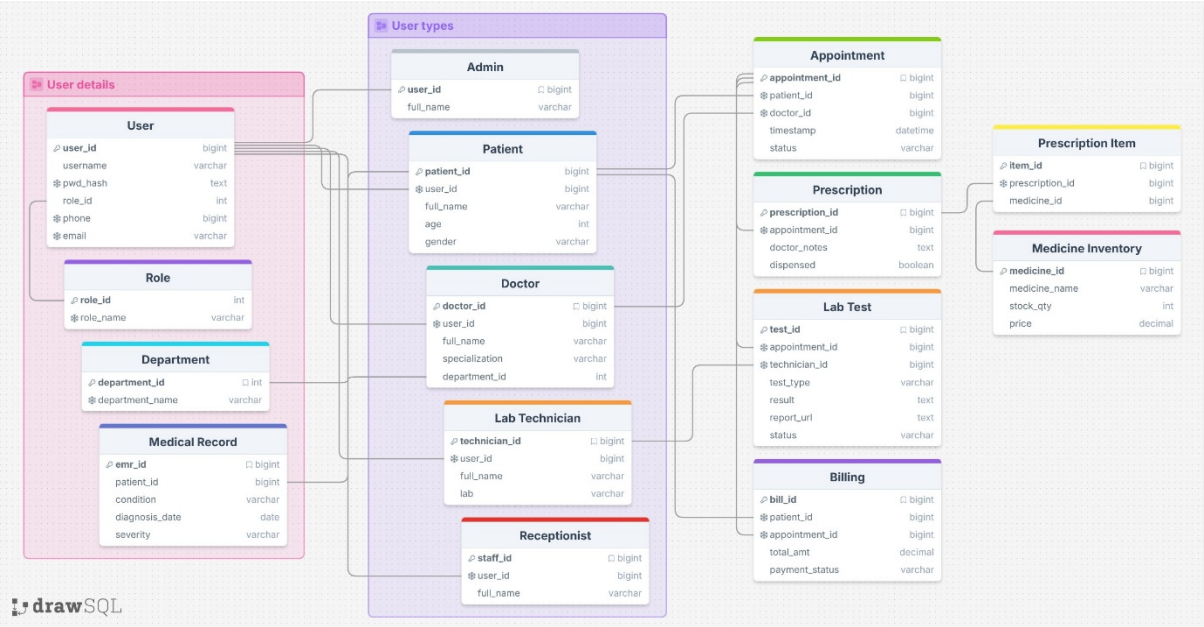
4.2 Relational Schema



4.3 Normalization

- 1NF (First Normal Form) - The tables are in 1NF as there are no multivalued or composite attributes. Each table contains atomic values, and all records are unique without any repeating groups. Therefore, the database satisfies the First Normal Form.

- 2NF (Second Normal Form) - As the database already satisfies 1NF, the next step is to ensure no partial dependencies exist. In all the tables, no non-prime attribute is dependent on a part of a composite primary key. Hence, the database satisfies the Second Normal Form.

- 3NF (Third Normal Form) - Since the database is already in 2NF, we need to ensure no transitive dependencies exist. There are no non-prime attributes that are transitively dependent on another non-prime attribute. Thus, the database satisfies the Third Normal Form.

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